

UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

COURSE: **MASTER OF SCIENCE IN DATA SCIENCE AND ANALYTICS**

COURSE UNIT: **DATA MINING MODELING AND ANALYTICS**

NAME: **KIWUMULO RITAH SHEEBA**

ACCESS NUMBER: **B35964**

ASSIGNMENT ONE PROJECT ONE REPORT

Data Exploration

The dataset contains 41,188 records and 21 columns: 20 input variables and the binary target y. Inputs describe customer demographics, financial attributes, contact and campaign history, and social and economic indicators. Customer age has a mean of approximately 40.02 years and ranges from 17 to 98 years. Initial inspection found no pandas null values. However, the accompanying README explains that the literal string 'unknown' is used to encode missing categorical attributes.

Target Distribution

The target variable is y hows substantial class imbalance. This means that accuracy alone is not an adequate measure of model usefulness; performance on the minority subscriber class must also be examined

Y (has the client subscribed a term deposit)	Number of records Records	Percentage
No	36,548	88.73%
Yes	4,640	11.27%

Unknown Categorical Values

The unknown fields which represent the missing fields include:

Variable	Unknown count
job	330
marital	80
education	1,731
default	8,597
housing	990
loan	990

2. Data Preparation and Justification

Preparation	Treatment and justification
Duplicate records	Removed 12 exact duplicate rows to avoid giving identical records unnecessary repeated weight.
Unknown categories	Retained as a separate category. The supplied README identifies 'unknown' as missing categorical information and allows it to be treated as a possible class label.
duration	Excluded because the README states that call duration is not known before a call and should be discarded for realistic pre-call prediction. Including it would cause target leakage.
pdays	Created previously_contacted = 1 when pdays is not 999, otherwise 0. .
Numerical missing values	Median imputation was placed inside the model pipeline and fitted on training data only.
Categorical variables	One-hot encoded within the pipeline; unseen test categories are ignored rather than causing errors.
Scaling	Numerical variables were standardised for Logistic Regression. Tree-based models did not require scaling.
Data splitting	Used stratified training, validation and test partitions. Stratification preserves the target-class proportions.

3. Majority-Class Baseline

A DummyClassifier using the most_frequent strategy was fitted on the training partition. It always predicts the majority class, 'no'. On the validation partition it achieved 88.74% accuracy, but precision, recall and F1-score for the subscriber class were all 0.00.

Baseline measure	Validation result
Accuracy	88.74%
Precision (subscriber class)	0.00%
Recall (subscriber class)	0.00%
F1-score (subscriber class)	0.00%

The baseline confusion matrix was TN = 5,846, FP = 0, FN = 742 and TP = 0. Although its accuracy is high, it fails to identify any potential subscribers. This demonstrates why accuracy alone is misleading for this imbalanced classification problem.

Model Selection and Ranking

Three classification approaches were trained and compared: Logistic Regression, Random Forest and HistGradientBoosting. Models were fitted on the training partition and compared on the validation partition at the default threshold of 0.50. PR-AUC (average precision) was prioritised because the subscriber class is the minority class. Precision, recall, F1-score and ROC-AUC were also considered.

Rank / Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
1. HistGradientBoosting	90.19%	67.91%	24.53%	36.04%	0.8182	0.4753
2. Logistic Regression	83.86%	37.27%	63.34%	46.93%	0.8067	0.4616
3. Random Forest	88.84%	50.48%	49.33%	49.90%	0.7970	0.4520

Justification of Ranking

1. HistGradientBoosting ranks first because it achieved the highest validation PR-AUC (0.4753) and ROC-AUC (0.8182), indicating the strongest overall ranking and discrimination among the tested models. It also achieved the highest precision at the default threshold, although its recall at 0.50 was low.

2. Logistic Regression ranks second by PR-AUC (0.4616). It provides an interpretable linear benchmark and achieved the highest recall at the default threshold (63.34%), making it useful when identifying a larger share of subscribers is important.

3. Random Forest ranks third by PR-AUC (0.4520). It captures nonlinear patterns and achieved the highest F1-score at the default threshold (0.4990), although its PR-AUC was lower than the other two models.

The ranking prioritises PR-AUC because the positive subscriber class is relatively rare. A different business objective, such as maximising recall or limiting call volume, could justify a different ranking.

Decision Threshold and Business Trade-off

The default probability threshold of 0.50 is not necessarily appropriate for campaign selection. The threshold was therefore selected using the validation data. Among the candidate thresholds tested, 0.25 maximised validation F1-score (0.5133), with validation precision of 48.56% and recall of 54.45%. The selected HistGradientBoosting model was then refitted on the complete training-plus-validation data and evaluated once on the held-out test set.

Held-out Test Results at Threshold 0.25

At threshold 0.25, 1,109 customers in the test set would be selected for calls. Of these, 540 were actual subscribers and 569 were non-subscribers (false positives). The model missed 388 of the 928 actual subscribers (false negatives).

Test result	Count / value
True negatives	6,739
False positives (unnecessary calls)	569
False negatives (missed subscribers)	388
True positives (subscribers identified)	540
Customers selected for calls	1,109
Precision (subscriber class)	48.69%
Recall (subscriber class)	58.19%
F1-score (subscriber class)	53.01%
Overall accuracy	88.38%

Explanation of the Trade-off

A lower threshold classifies more customers as likely subscribers. This generally increases recall and reduces missed subscribers, but also increases false positives and unnecessary marketing calls. A higher threshold reduces false positives and campaign workload, but increases false negatives and missed subscription opportunities.

Recommendation: use 0.25 as an initial operating threshold if the bank seeks a balance between precision and recall. It is not a universal optimum. The final operating threshold should reflect the cost of a call, expected revenue from a subscription and available campaign capacity. If calls are expensive or capacity is limited, increase the threshold. If missing potential subscribers is more costly, lower the threshold.

Conclusion

The Bank Marketing dataset is imbalanced, and the majority-class baseline's 88.74% validation accuracy does not indicate useful subscriber identification. HistGradientBoosting achieved the strongest validation PR-AUC among the three evaluated approaches. Selecting a threshold of 0.25 using validation F1-score produced 58.19% recall and 48.69% precision on the held-out test set. The recommended approach is therefore HistGradientBoosting with an initial threshold of 0.25, subject to adjustment based on the bank's marketing costs and call capacity.